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DR. TOBIAS LEVA

POSTDOCTORAL RESEARCHER

EDUCATION

PHD SYSTEMS NEUROSCIENCE

magna cum laude
May 2017 – Feb. 2024
Humboldt University, Berlin
3 year fellowship (top 1%; 10/1000)

DIPLOMA BIOPHYSICS

Grade: 1.1 (Range: 1.0 - 4.0)
Oct. 2009 – March 2015
Goethe University, Frankfurt

CERTIFICATES

TENSORFLOW DEVELOPER

Machine Learning, CNN, NLP,
Sequences, Time Series and Prediction
Deeplearning.ai

DATA SCIENTIST

Python, Unix, SQL, Git, GitHub
EDA, Statistics, ML algorithms,
ML Engineering, Model Evaluation,
Performance metrics
neueFische GmbH

PUBLICATIONS

The spatial representation of temperature in the thalamus.
Leva, Whitmire, Poulet
Neuron (in revision) (2024)

The spatial investigation of temperature across the thalamus.
Leva
PhD Thesis (2024)

Thermosensory thalamus: parallel processing across model organisms.
Leva, Whitmire
Frontiers in Neuroscience (2023)

Brain-wide connectivity map of mouse thermosensory cortices.
Bokiniec, Whitmire, **Leva**, Poulet
Cerebral cortex (2023)

PROFILE

I am a systems neuroscientist with 5+ years of hands-on experience in high-density extracellular recordings (Neuropixels, Imec), in code development of signal processing models and applying machine learning and statistical techniques to high-dimensional neural data, as demonstrated by my scientific work and certifications in data science and neural networks. I possess extensive expertise in planning and conducting innovative neuroscience research, data analysis, and visualization, as well as in preparing scientific manuscripts. This is demonstrated by two peer-reviewed publications and one preprint resulting from my doctoral work. My professional approach emphasizes the translation and transfer of scientific knowledge through collaboration, training, and effective communication, as demonstrated through various teaching roles and mentorship responsibilities.

CURRENT POSITION

Postdoctoral Researcher — JULY 2024 - PRESENT
Neural Circuits and Behavior, MDC Berlin

- Developed signal processing and statistical models for high-dimensional time series data of animal behavior during temperature perception task
- Established method for manipulation of neural activity including design and prototyping of custom device and control software development
- Created a comprehensive Python end-to-end analysis pipeline for data acquisition, preprocessing, signal detection, statistical analysis, and data visualization
- Led a multidisciplinary team of scientists, engineers, and technicians, managing project progress, training and mentoring and fostering a collaborative, results-driven lab culture

ACADEMIC EXPERIENCE

Doctoral Researcher — MAY 2017 - FEB 2024
Neural Circuits and Behavior, MDC Berlin

- Independently designed, planed and executed experimental large-scale systems neuroscience study to determine the neural basis of temperature perception, resulting in two peer-reviewed publications and one preprint as well as presentation at scientific conferences.
- Developed a GitHub-shared data processing pipeline for high-dimensional time series dataset (>10 TB) of neural activity, covering the entire data science life cycle
- Designed and engineered a cutting-edge high-density extracellular electrophysiology setup to record single unit temperature-evoked neural activity, integrating various real-time sensor data, hardware and software development for data acquisition and big data management
- Instructed 'Data Analysis in Python' course within the Master's curriculum over 2 terms and mentored PhD and Master's students
- Founded and organized the "Circuits Club" neuroscience seminar series
- Secured independent research funding, one of ten recipients selected from 1000 applicants

HARD SKILLS

- Python
- Time series analysis, Statistics
- Machine learning, Neural networks
- Data visualization
- Code development
- Version Control
- Database systems
- Hardware/software - interfacing
- CAD, 3D printing, Laser-cutting

SOFT SKILLS

- Analytical thinking
- Problem-solving
- Excellent communication German, English
- Project management
- Team player
- Teaching, Mentoring

TEACHING

Data analysis in Python
group of +20 Master's students
Einstein Center for Neuroscience
Charité, Berlin (2020 - 2022)

Principles of metabolism
group of +10 Bachelor's students
Institute for Biochemistry
Goethe University, Frankfurt (2012)

Kinetics and electrostatics
group of +10 Bachelor's students
Institute for Biophysical Chemistry
Goethe University, Frankfurt (2012)

REFERENCES

Prof. Dr. James Poulet
Group Leader *MDC Berlin*
james.poulet@mdc-berlin.de

Dr. Clarissa Whitmire
Group Leader *Queensland Brain Institute*
jc.whitmire@uq.edu.au

Graduate Student — MAY 2014 - MAY 2015

Max-Planck Institute for Biophysics, Frankfurt

- Applied biophysical models to investigate functionality and dynamics of genetically encoded membrane voltage sensors
- Performed electrophysiological recordings and managed diverse bio-physical, genetic, microbiological, and biochemical laboratory tasks daily
- Received intensive training in quantitative methods (physics, statistics, mathematics) through cross-domain collaboration and supervision

PROFESSIONAL EXPERIENCE

Founder & Management — AUG 2015 - FEB 2016

'Badias' Restaurant, Schirn Kunsthalle, Frankfurt

- Co-founded and managed restaurant
- Recruited and led a diverse team of employees
- Oversaw time management, budgeting, strategic placement, goal-setting
- workflow optimization to ensure efficient operations

VOLUNTARY RESEARCH INTERNSHIPS

Research Assistant — MAY 2016 - OCT 2016

Sensory Systems & Behavior, CRG Barcelona

- Executed cutting-edge systems neuroscience research
- Confocal and two-photon microscopy, manipulations of neural activity via Optogenetics, high- speed video recordings for locomotion tracking
- Extraction of movement features from videos and analysis via custom Matlab code

Research Assistant — JAN 2013 - NOV 2013

MPI Human Cognitive Brain Sciences, Leipzig

- Collaboratively managed a large-scale clinical and functional imaging study investigating the effects of meditation on higher brain functions
- Part of diverse team from different scientific background
- Conducted behavioral experiments with human participants
- organizing and overseeing weekly testing and training sessions for experimental groups of over 40 individuals over several months

CONFERENCE PRESENTATIONS

GRC Thalamocortical Interactions — 2022

Distinct Cellular Encoding of Warm and Cool in the Thalamus
Lucca, IT

35th Annual Barrels Meeting — 2022

Distinct Cellular Encoding of Warm and Cool in the Thalamus
La Jolla, US